HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2014)

Student No.: 15 Name: _____ Marks: / 38

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. There are 792 workers in a factory. If the number of male workers is 20% less than that of female workers, then the number of male workers is

   A. 352.

   B. 360.

   C. 432.

   D. 440.

(DSE-2014 PII Q09)

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2. If the angle and the radius of a sector are decreased by $x\%$ and 50% respectively so that its area is decreased by 90%, then $x =$

A. 20.

B. 40.

C. 60.

D. 80.

(DSE-2014 PII Q10)

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3. It is given that $\frac{4}{5a} = \frac{5}{7b} = \frac{7}{9c}$, where a, b and c are positive numbers. Which of the following is true?

A. $a < b < c$

B. $a < c < b$

C. $b < a < c$

D. $b < c < a$

(DSE-2014 PII Q12)

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4. Let a_n be the nth term of a sequence. If $a_2 = 7$, $a_4 = 63$ and $a_{n+2} = a_{n+1} + a_n$ for any positive integer n, then $a_5 =$

A. 56.

B. 70.

C. 91.

D. 119.

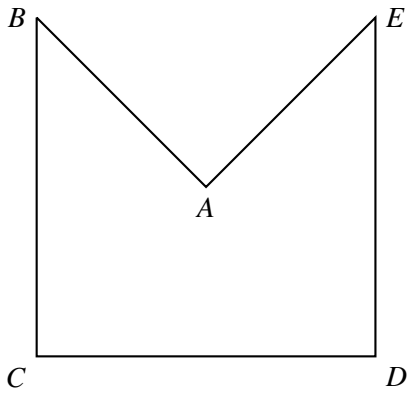
(DSE-2014 PII Q14)

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5. In the figure, $AB = AE$ and $\angle BAE = \angle BCD = \angle CDE = 90°$. If $BC = CD = DE = 16$ cm, then the area of the pentagon $ABCDE$ is



A. 71 cm^2.

B. 128 cm^2.

C. 192 cm^2.

D. 224 cm^2.

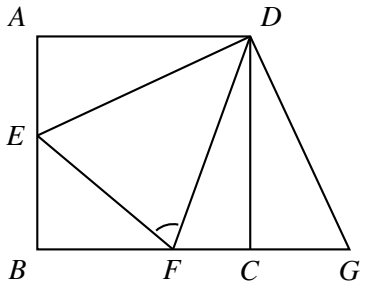
(DSE-2014 PII Q15)

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6. In the figure, $ABCD$ is a square. BC is produced to G such that $\angle CDG = 25°$. E is a point lying on AB such that $AE = CG$. If F is a point lying on BC such that $\angle CDF = 20°$, then $\angle DFE =$



A. 60°.

B. 65°.

C. 70°.

D. 73°.

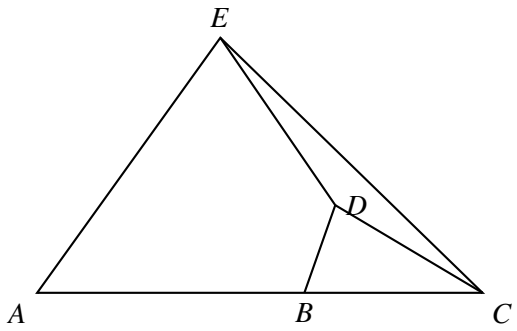
(DSE-2014 PII Q16)

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7. In the figure, B is a point lying on AC such that $AB : BC = 3 : 2$. It is given that $AE \parallel BD$. If the area of $\triangle BCD$ and the area of $\triangle CDE$ are 4 cm^2 and 8 cm^2 respectively, then the area of the trapezium $ABDE$ is



A. 18 cm^2.

B. 21 cm^2.

C. 27 cm^2.

D. 33 cm^2.

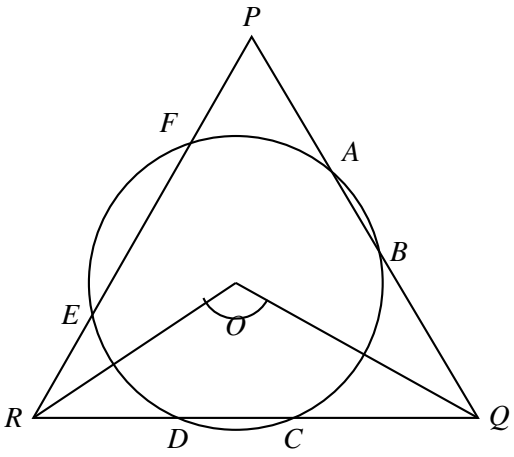
(DSE-2014 PII Q17)

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8. In the figure, O is the centre of the circle $ABCDEF$. $\triangle PQR$ intersects the circle at A, B, C, D, E and F. If $\angle QPR = 38^\circ$ and $AB = CD = EF$, then $\angle QOR =$



A. 109°.

B. 117°.

C. 123°.

D. 142°.

(DSE-2014 PII Q21)

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9. The equations of the straight lines L_1 and L_2 are $2x+3y=5$ and $4x+6y=7$ respectively. If P is a moving point in the rectangular coordinate plane such that the perpendicular distance from P to L_1 is equal to the perpendicular distance from P to L_2, then the locus of P is a

A. circle.

B. square.

C. parabola.

D. straight line.

(DSE-2014 PII Q24)

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10. A box contains m yellow balls and 20 black balls. If a ball is randomly drawn from the box, then the probability of drawing a yellow ball is $\frac{1}{m}$. Find the value of m.

A. 4

B. 5

C. 15

D. 25

(DSE-2014 PII Q27)

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11. The mean height of 25 teachers and 140 students is 150 cm. If the mean height of the students is 145 cm, then the mean height of the teachers is

A. 151 cm.

B. 155 cm.

C. 176 cm.

D. 178 cm.

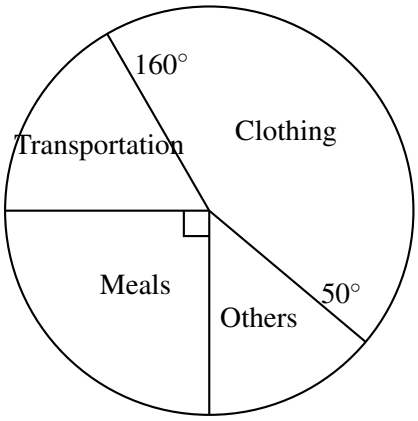
(DSE-2014 PII Q28)

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12. The pie chart below shows the expenditure of John in a certain week. John spends $240 on clothing that week. Find his expenditure on transportation that week.



A. $40

B. $60

C. $90

D. $135

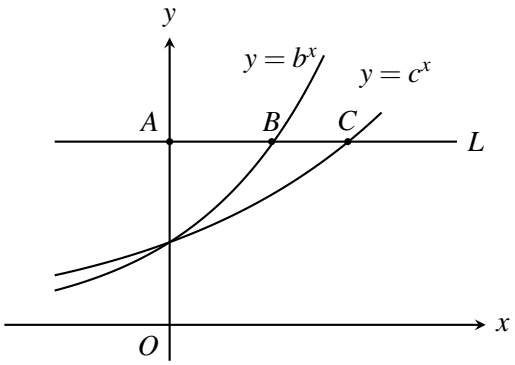
(DSE-2014 PII Q29)

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13. The figure shows the graph of $y = b^x$ and the graph of $y = c^x$ on the same rectangular coordinate system, where b and c are positive constants. If a horizontal line L cuts the y-axis, the graph of $y = b^x$ and the graph of $y = c^x$ at A, B and C respectively, which of the following are true?



I. $b < c$

II. $bc > 1$

III. $\dfrac{AB}{AC} = \log_b c$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2014 PII Q32)

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14. Which of the following is the greatest?

A. 124^{241}

B. 241^{214}

C. 412^{142}

D. 421^{124}

(DSE-2014 PII Q33)

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15. $7 \times 2^{10} + 2^8 + 5 \times 2^3 - 2^3 =$

A. 111010100000_2.

B. 111100010000_2.

C. 1110100100000_2.

D. 1111000010000_2.

(DSE-2014 PII Q34)

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16. If $m > 1$, which of the following are geometric sequences?

I. $2^m, 2^{2m}, 2^{3m}, 2^{4m}$

II. $m, 2m^2, 3m^4, 4m^8$

III. $\log m, \log m^2, \log m^4, \log m^8$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2014 PII Q37)

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17. For $0° \leq x \leq 360°$, how many roots does the equation $7\sin^2 x = \sin x$ have?

A. 2

B. 3

C. 4

D. 5

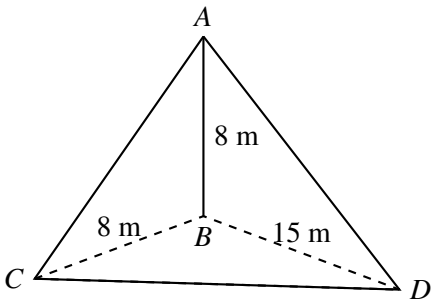
(DSE-2014 PII Q39)

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18. In the figure, AB is a vertical pole standing on the horizontal ground BCD, where $\angle CBD = 90°$. If the angle between the plane ACD and the horizontal ground is θ, then $\tan\theta =$



A. $\frac{8}{15}$.

B. $\frac{15}{8}$.

C. $\frac{15}{17}$.

D. $\frac{17}{15}$.

(DSE-2014 PII Q40)

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19. In an examination, Peter gets 55 marks and his standard score is -3 while Mary gets 95 marks and her standard score is 2. Find the mean of the examination scores.

A. 8 marks

B. 64 marks

C. 75 marks

D. 79 marks

(DSE-2014 PII Q44)

I can finish _____ questions

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